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Coursework Brief
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C2O — Close-to-Open
Cross-Sectional Overnight Strategy
Author of the brief
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Coursework Brief

C2O — Close-to-Open

A cross-sectional, dollar-neutral overnight strategy on US large-cap equities

Group coursework — groups of up to 3 students

Course title

Machine Learning in Finance

Author of the brief

Dr Arnaud de Servigny

This document specifies the assignment, the data you may use, and the deliverables. It does not specify a method.

Executive summary — what this coursework is about

This document is the brief for your group coursework. It tells you, plainly, what you are being asked to build, what data you may use, what constraints you must respect, and how your work will be marked. It does not tell you how to solve the problem. The how is your job, and the marks reflect that.

The starting point is a simple empirical observation. If you take a major US stock index over the past two decades and split each day's return into two pieces — the part earned overnight, between yesterday's closing price and today's opening price, and the part earned during the trading day, between today's open and today's close — almost the entire long-run return is contained in the overnight piece. The intraday piece is approximately flat or slightly negative depending on the period. Section 1 shows this picture.

Your task is to take that index-level fact and turn it into a tradable strategy on individual stocks. Concretely, you will design and back-test a daily-frequency, dollar-neutral, US large-cap equity strategy that holds positions only overnight: long stocks bought at today's closing auction and sold at tomorrow's opening auction; short stocks sold at today's closing auction and bought back at tomorrow's opening auction. Every position is held for one overnight session, no longer. The portfolio is rebuilt every trading day.

Five steps structure your work, and each is a section of this brief:

1. Build a clean daily panel of US equities and define the return objects you will work with.
2. Choose a tradable universe — at the daily horizon, with realistic capacity constraints.
3. Filter the short leg for hard-to-borrow names, since their financing costs can erase the alpha.
4. Construct a cross-sectional alpha that ranks the eligible stocks for tomorrow's overnight return. This is the heart of the work and the part where we expect to see your creativity.
5. Construct a market-neutral portfolio from that ranking, account honestly for trading costs, and report performance in a defensible way.

There is one important constraint up front. You will work only from daily-frequency data: end-of-day open, high, low, close, volume, daily fundamentals, daily corporate-action and earnings flags, and bi-monthly short-interest releases. No intraday quotes, no tick data, no order-book information, no news feeds at sub-daily granularity. This restriction is part of the assignment, not a limitation we apologise for. Many overnight strategies that look attractive in published work rely on microstructure inputs that you would not have on a deployable desk; we want to see what you can build under the discipline of daily-only inputs.

What we are looking for, summarised in one sentence: a defensible answer that you understand. The headline number you target — a net Sharpe of around 1.0 against realistic frictions — is aspirational, not a pass/fail threshold. We do not expect every group to meet it, and we have no intention of grading by Sharpe alone. We are looking for a strategy where every step from raw data to final number is traceable, justified, and free of look-ahead bias, and where you can articulate honestly what your reported number represents. An honest 0.6 with a clean methodology and a clear-eyed diagnosis of why the strategy is not stronger will mark above an optimistic 2.5 that fails the audit, every time.

How to read this brief

Each section poses questions and gives you the data and constraints needed to answer them. We deliberately do not propose specific algorithms, feature sets, or model architectures — those are choices you must make and justify.

Where you see a numbered question in a section, treat it as a checkpoint: your final report should make explicit how you addressed it.

If a choice we leave open seems important and under-specified, that is intentional. The decision, and your reasoning behind it, is part of what is being graded.

Contents

1. Context — the empirical puzzle behind the strategy

1.1 The stylised fact

Consider the cumulative return of a major US equity index — the S&P 500 or the Russell 1000 — over the past two decades. Decompose each daily return into two non-overlapping components: the overnight return, earned from the previous day's closing price to the current day's opening price; and the intraday return, earned from the current day's opening price to its closing price. By construction, compounding the two together reproduces the full close-to-close return.

When you carry out this decomposition empirically, you observe something that is at first counter-intuitive: the long-run cumulative return of the index is almost entirely contained in the overnight component. The intraday component is approximately flat or slightly negative, depending on the sub-period. The overnight return shows lower volatility per unit of time than the intraday component, which means that the overnight Sharpe ratio of being long the index is materially higher than the intraday Sharpe ratio. Figure 1.1 illustrates the shape with synthetic but representative numbers.

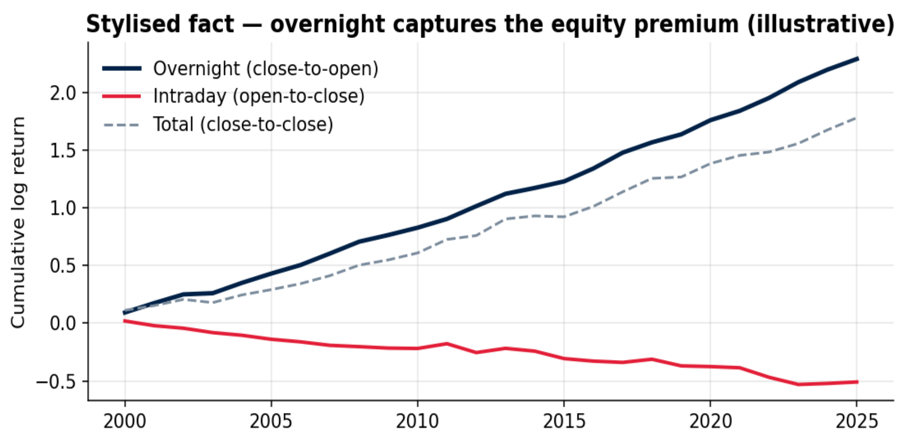

Figure 1.1 — Indicative shape of the cumulative-return decomposition for a US large-cap index. Solid: overnight component, compounded. Dashed: total close-to-close. Red: intraday component. Your first analytical task is to reproduce this picture with your own data.

1.2 Why a single-stock cross-sectional strategy is not the same problem

Index-level evidence does not translate directly into a profitable strategy on individual stocks. There are at least three reasons you should think about before designing anything:

- Per-stock execution requires two auctions every day — a closing auction at end of day t and an opening auction at start of day t+1 — and each leg is subject to slippage, financing, and exchange fees. The round-trip cost is amplified by the daily turnover of the strategy.
- Going short requires locating shares to borrow. For some stocks this is essentially free; for others the borrow fee is comparable in magnitude to the alpha you hope to harvest.
- The cross-section of overnight returns is dominated by stock-specific events — earnings, guidance changes, corporate actions, index inclusions — that have to be handled deliberately. They are not noise; they are signal sometimes, and contamination at other times.

Your strategy will live or die in the way it handles those three realities, not in the elegance of any predictive model.

1.3 What the brief asks for, in one paragraph

Build a daily, dollar-neutral, long–short equity strategy on US large-cap stocks that captures part of the cross-sectional dispersion of overnight returns. Long positions are entered at the closing auction at the end of trading day t and exited at the opening auction at the start of trading day t+1. Short positions follow the same schedule. As an aspirational headline, we suggest aiming for a net Sharpe ratio of order 1.0 after honest accounting for trading costs and short-borrow financing — but this is a target you should think of as something to push toward, not as a threshold for passing the coursework.

Development window and held-out evaluation

You will develop, train, validate and back-test your strategy strictly on data from the start of 2010 through the end of 2024. That is your full available window. No part of your modelling pipeline — feature construction, hyper-parameter choice, threshold calibration, regime tuning, or anything else — may use information from 2025 or later.

Your submitted code will be re-run by the marker on a held-out window covering 2025 and 2026. The performance of the strategy on that held-out window is part of how your work is graded. Treat 2025–2026 as if it does not exist while you are designing the strategy.

If your code reads any data dated after 31 December 2024 in the development phase — even passively, even for plotting — you have leaked the held-out window into your design and the corresponding submission is invalid. Build your data loader so that the cutoff is a single configurable parameter, set to 2024-12-31 by default.

2. Step 1 · Building the daily panel

2.1 The return objects

Your first job is to define, precisely, the quantities you will be predicting and trading. Adopt the following notation throughout your work: let day t denote a trading day; let Open_t and Close_t denote the official opening and closing auction prices on day t, adjusted for splits and dividends; and let Volume_t denote the consolidated daily share volume.

The overnight return on day t is the return earned from the closing auction at the end of day t−1 to the opening auction at the start of day t:

$$r_{ON,\,t} \;=\; \left(\, Open_t \;\div\; Close_{t-1} \,\right) \;-\; 1$$

The intraday return on day t is the return earned from the opening auction to the closing auction on the same day:

$$r_{ID,\,t} \;=\; \left(\, Close_t \;\div\; Open_t \,\right) \;-\; 1$$

The two pieces compound to the conventional close-to-close return:

$$\left(\, 1 \;+\; r_{ON,\,t} \,\right) \times \left(\, 1 \;+\; r_{ID,\,t} \,\right) \;-\; 1 \;=\; \left(Close_t \;\div\; Close_{t-1}\right) \;-\; 1$$

All three numbers must reconcile to within a numerical tolerance you will define and report. Any residual that does not reconcile is the symptom of a data-cleaning issue — typically a corporate action handled inconsistently between the open price, the close price, and the volume — and it must be diagnosed, not silently masked.

2.1.1 Auction conventions and the trader's clock

Two conventions matter for everything that follows. A Market-on-Close (MOC) order is an order that executes only in the closing auction at the official exchange close, which on US exchanges is 16:00 Eastern Time. A Market-on-Open (MOO) order is an order that executes only in the opening auction at the official exchange open, which is 09:30 Eastern Time. In this brief, both legs of every position execute at auction prices: long entries and short entries via MOC at day t; long exits and short covers via MOO at day t+1.

The relevant decision time for every trade is therefore the late afternoon of day t, before the 16:00 ET closing auction. A common convention is to assume orders are submitted at 15:50 ET. Any feature your alpha uses must be observable by 15:50 ET on day t — that is, derivable from data closed on day t−1 or from intraday data on day t that has already been published before 15:50 ET. Daily features built from Close_t are not observable at decision time and cannot be used without a one-day shift.

2.1.2 Earnings dates and the timing flag

Earnings announcements are reported with a timing flag that you must respect. A Before Market Open (BMO) announcement is released before the 09:30 ET open of the day it is dated. An After Market Close (AMC) announcement is released after the 16:00 ET close of the day it is dated. A Same-Day announcement is released during regular trading hours.

This matters because an AMC announcement on day D is unknown to a trader placing MOC orders at 15:50 ET on day D — the news is, by definition, published only after MOC orders have already been entered. Any feature that conditions on an AMC earnings event on day D is using future information when used to predict r_ON, D+1. The fix is to shift the effective announcement date forward by one trading day for AMC events. A BMO announcement on day D, by contrast, is observable by 09:30 ET that day and is therefore available to a trader placing MOC orders the same afternoon.

2.1.3 Short interest and its publication lag

Short-interest data is released by FINRA on a bi-monthly cycle. Settlement-date snapshots dated on the 15th and the last business day of each month are published approximately eight calendar days later. For your features, you must apply two additional days of lag on top of the official publication date to reflect realistic data-vendor delivery. A snapshot dated D, officially published on date D+8, is treated as available in your features only from D+10 onwards.

Concretely: build a daily-frequency short-interest column where the value on date D is the most recent published snapshot whose effective availability date is on or before D−1. Forward-fill that column between releases. The same convention applies to any short-interest-derived quantity (days-to-cover, change-in-short-interest, etc.).

2.1.4 Corporate actions, survivorship, and the universe at the time

Your panel must be survivorship-bias-free. Every stock that was in the eligible universe at any historical point must remain in the panel through to the date of its delisting, with returns reflecting realistic delisting outcomes (acquisition payouts, bankruptcy stubs, etc.). Splits, dividends, special dividends, spin-offs, and rights issues must be applied consistently to prices and volumes. A panel that drops names retroactively after a delisting will overstate returns; correcting this is your responsibility, not optional.

2.2 The investment universe — chosen yearly, held fixed within the year

Define the universe of eligible stocks as follows. On the first trading day of each calendar year Y, rank all listed US common equities by market capitalisation as of the last trading day of year Y−1, restricting to those with at least 12 months of price history. Take the top 1 000 names by that ranking and freeze that list as the eligible universe for the entire calendar year Y. Do not re-rank, add, or drop names within a calendar year except via delisting events (which are handled mechanically: a delisted name leaves the universe on its delisting date with a documented payout).

This rule has three consequences you should be aware of. First, the universe is point-in-time: the list of names eligible on any historical date is exactly the list a trader could have constructed at the start of that year using only past information. Second, mid-year IPOs and spin-offs are not in the universe in the year of their issuance — they become eligible only at the next 1-January cut. Third, you may legitimately tighten the universe further at decision time using filters described in Section 3 (capacity, price, volume, earnings windows). The 1 000-name list is the floor of eligibility, not the trading set.

2.3 Sanity checks before you start modelling

Before any signal construction, verify on your panel that the index-level stylised fact of Section 1 is reproduced. Construct an equally-weighted portfolio of the names eligible in your universe on each date, and compute its overnight component, intraday component, and total close-to-close return. Plot the cumulative growth of $1 invested in each of the three streams. The overnight stream should dominate the total stream over the long run, and the intraday stream should be flat or negative. If your numbers do not show this shape, your data is wrong, and any modelling on top of it will be uninformative.

Document this verification step. The marker will look at it before reading anything else.

Section 2 — questions you must explicitly answer in your report

1. What is your data source, what is its window, and what does its survivorship correction look like?
2. How do you reconcile open, close, and volume after corporate actions? What is your residual tolerance and what fraction of stock-days fail it?
3. How do you encode the AMC / BMO timing flag for earnings? Show the rule and verify it on at least one stock-event you cite by hand.
4. What is your short-interest construction and the precise lag applied? Plot the resulting series for a representative name.
5. How does your eligible universe evolve year by year? Provide a count of eligible names at each year-start and a count of mid-year exits.
6. Does the equal-weighted portfolio of your eligible universe reproduce the stylised fact of Section 1.1? Show the figure and quantify the dispersion across years.

3. Step 2 · Defining a capacity-aware universe

3.1 The capacity question, framed correctly

The first question students typically ask about a stock is "can I trade it?". For US large-cap, the answer is almost always yes for a small ticket. The harder and more relevant question is "can I trade it in the size implied by my assumed capital, at the closing auction, without paying away the alpha I expect to harvest?". That is a per-stock, per-date, per-AUM question, and answering it requires three quantities: a measure of the usable liquidity of the stock; a model that translates a target trade size into expected slippage; and an explicit assumption on the dollar capital you intend to commit.

3.2 Liquidity proxies you have available

Without intraday data, your usable proxies are coarse. Three are conventional and sufficient for this brief:

- Average Daily Volume over a rolling window (call it ADV20 — the trailing 20-day mean of daily share volume × close price, expressed in dollars). This is your single best per-stock liquidity number.
- A rolling effective-spread estimator built from daily prices. The Roll spread is one well-known choice; there are others. A daily-frequency spread proxy will be noisy on individual stocks but is informative as a cross-sectional ranking signal.
- Price per share. Penny stocks behave pathologically because the relative tick size becomes large, so a hard floor on price per share is a clean and defensible filter.

3.3 A simple market-impact convention

Adopt the conventional square-root impact model that the practitioner literature has converged on. For a child order of dollar size Q sized as a fraction f of the daily volume the stock typically trades, the expected slippage in basis points is approximately

$$Impact \approx \mathrm{k} \times \sigma_{daily} \times \sqrt{f} \times 10^4$$

where σ_daily is the stock's daily realised volatility and k is a venue-specific constant in the range 0.5 to 1.0 documented in the practitioner literature. For closing auctions, the effective k is lower than for continuous trading because participation as a fraction of auction-only volume is materially higher, but auction volume is itself smaller; the two effects roughly offset for a first-pass estimate. You may take k = 0.7 for the closing auction unless you can defend a different number.

With this in hand, your task is to choose a participation-rate cap — what is the maximum fraction f you will permit any single position to be of the stock's daily ADV — and to verify that the implied slippage at that cap is acceptable relative to your expected per-stock alpha.

3.4 Capacity by AUM — distinguishing per-stock and portfolio AUM

Two distinct dollar quantities appear in any capacity discussion and must not be conflated.

- Portfolio AUM — the total dollar capital committed to the strategy. This is the number you would quote to an investor. For this coursework you will report results at three explicit portfolio-AUM levels: 50 million dollars, 250 million dollars, and 1 billion dollars.

- Per-stock AUM — the dollar value of the position taken in any single name on any single date. This is bounded mechanically by the portfolio AUM, by the number of names you decide to hold simultaneously (long and short), and by your participation cap as a fraction of that name's ADV. The binding constraint at the per-stock level is typically the participation cap, not the portfolio share.

A worked numerical example clarifies. Suppose your portfolio AUM is 250 million dollars, you intend to hold roughly 100 long names and 100 short names with equal dollar weights, and you cap participation at 5 percent of ADV. The maximum nominal per-stock position is 250 million ÷ 200 = 1.25 million dollars on each side. If a candidate name has an ADV of 10 million dollars, the participation cap allows up to 0.5 million dollars per side, which is binding: the position is sized at 0.5 million, not 1.25 million, and the residual capital must be redistributed across the rest of the basket. If a candidate name has an ADV of 50 million dollars, the position is sized at the equal-weight 1.25 million and participation is 2.5 percent. Your portfolio construction logic must implement this distinction explicitly.

3.5 Output of Step 2 — the eligibility table

The deliverable from Section 3 is a per-(stock, date) eligibility table with the binding constraint recorded for each row. The eligibility column should take values such as ADV_FAIL (failed the dollar-volume floor), MCAP_FAIL (failed the market-capitalisation floor at year-start), PRICE_FAIL (price below the per-share floor), VOL_FAIL (annualised realised volatility outside the chosen band), EARN_WINDOW (within an earnings exclusion window — your choice of width), or OK. Any name not flagged OK is excluded from the trading set on that date.

You will report the binding-constraint distribution across years. In some sub-periods (the first half of 2020 in particular) a meaningful share of the universe drops out on the volatility floor; documenting this is part of explaining your performance in those windows.

> **Section 3 — questions you must explicitly answer in your report**
>
> 1. What thresholds do you set on dollar volume, market capitalisation at year-start, price per share, realised volatility, and the earnings window? Justify each numerically.
> 2. What participation cap do you assume in the closing auction, and what is the implied slippage at that cap for a typical mid-cap and a typical large-cap name in your universe?
> 3. How does the eligible set evolve through 2010–2024 at each of the three portfolio-AUM levels you must report (50M, 250M, 1B)?
> 4. What is the binding-constraint distribution? Provide a table by year of how many stock-days are excluded by each reason.

4. Step 3 · Filtering the short leg for borrow cost

4.1 Why the short leg needs separate treatment

The long leg of a US large-cap strategy faces no fundamental access constraint: you can buy any listed stock at the closing auction without seeking permission. The short leg is qualitatively different. A short sale requires locating shares to borrow from a holder willing to lend them, against a fee. For most large-cap stocks at most points in time, the borrow market is deep and the fee is small (the so-called General Collateral, or GC, rate, of order 25 to 50 basis points per annum). For a meaningful minority of names in your universe, however, the borrow is expensive — fees of 100 to several hundred basis points per annum — and for a small minority it is essentially unavailable at any price.

Expensive borrow is not a tail risk you can ignore. It is concentrated in specific economic situations: stocks with high short interest relative to float; stocks with low public float relative to market capitalisation; stocks experiencing distress or short-squeeze dynamics. These same situations are often where a naive momentum-reversal model produces its strongest short signals. If your alpha concentrates the short book in expensive-to-borrow names, the financing cost can plausibly exceed the alpha itself.

4.2 What you have to identify hard-to-borrow names

You do not have direct access to a real-time hard-to-borrow indicator from a prime broker. You must build a proxy from the daily data described in Section 2. The available inputs include:

- Reported short interest as a fraction of shares outstanding.
- Days-to-cover, defined as reported short interest divided by some rolling-average daily volume.
- Realised borrow stress signals — large jumps in short interest, or persistent rises across consecutive bi-monthly snapshots.
- Cross-sectional flags such as small market capitalisation conditional on a given short-interest level, or unusually high short-interest in a stock that has a recent IPO or a depressed float.

How you combine these inputs is your decision. Whatever rule you adopt, it must produce a per-(stock, date) flag that is point-in-time given the lag conventions of Section 2.1.3.

4.3 Modelling borrow as a cost, not as a hard exclusion

There are two clean ways to reflect borrow in your back-test. You may adopt either or a combination, but you must justify the choice.

1. Hard exclusion — the short side cannot enter any name flagged hard-to-borrow. The flag becomes part of the eligibility table of Section 3.5. This is conservative and easy to implement, but it can throw away genuine alpha when the flag misfires.
2. Tiered cost — every short position is charged a daily-equivalent borrow fee whose magnitude depends on a tier assigned to the stock at decision time. The effect is to reduce the realised return of the short leg by the fee. Section 6 specifies a precise three-tier schedule that you must use for the headline results.

Whichever approach you take, the headline numbers in your final report must be net of borrow cost calculated under the schedule given in Section 6. A back-test that ignores borrow does not constitute a valid submission.

Section 4 — questions you must explicitly answer in your report

1. What is your hard-to-borrow proxy? Specify the inputs, the combination rule, and the threshold.
2. How does your proxy validate against any external evidence you can access (occasional vendor data, journalistic reports of short squeezes, etc.)? Even one or two anecdotes documented in your report are useful.
3. Do you adopt hard exclusion, tiered cost, or both? What fraction of your raw short signal is affected by your treatment?
4. Quantify the impact: what is the gross-of-borrow Sharpe and the net-of-borrow Sharpe under the cost schedule of Section 6?

5. Step 4 · Constructing the cross-sectional alpha

5.1 The prediction problem, stated

At decision time on day t — that is, at 15:50 ET on day t, before the closing auction — you must produce, for each stock i in the eligible universe of that date, a real-valued score s_{i,t}. The score must rank the cross-section in order of expected overnight return for the next session. Higher scores mean larger expected r_ON, t+1; lower scores mean smaller or more negative expected r_ON, t+1. The portfolio-construction step in Section 6 will turn these scores into long and short positions; the quality of the strategy is bounded above by the quality of the scores.

Formally, your alpha is a function s that takes as input the information set available to a trader at 15:50 ET on day t — call it $\mathcal{F}$_t — and returns a vector of scores indexed over the eligible universe:

$$s_{i,t} \;=\; s\,(\,\mathcal{F}_t\,;\,\theta\,) \qquad \text{for } i \in \mathcal{U}_t$$

where $\mathcal{U}$_t is the eligible universe on date t and θ are the parameters of whatever model you choose. The alpha is evaluated by the cross-sectional rank correlation between s_{i,t} and the realised r_ON, i, t+1 — i.e. the Information Coefficient, computed daily and averaged over time. A persistently positive Information Coefficient is the necessary condition for a profitable strategy.

5.2 What this section deliberately does not specify

This section is the most open-ended of the brief. It does not prescribe a model class, a feature set, a training procedure, or a target transformation. Those are choices you must make and justify, and they are where the marks differentiate. We are deliberately silent on the following:

- Whether you use a linear model, a tree-based model, a neural network, a heuristic ranking, or an ensemble. All are admissible. None is preferred a priori.
- What your features should be. Anything observable in $\mathcal{F}$_t and reproducible in real time is allowed: lagged returns, realised volatility, valuation ratios from fundamentals, technical indicators, sentiment proxies derived from corporate-event flags, post-earnings drift signals, calendar effects, factor-loading estimates. The choice — and the discipline of point-in-time construction — is yours.
- What target you fit to. Raw r_ON, t+1 is the obvious choice, but a winsorised version, a residualised version (e.g. r_ON, t+1 minus the cross-sectional mean of the day, or minus a factor-model expectation), or a transformed version (rank, sign, decile) is equally legitimate and may be preferable. Justify your choice.
- How you train and validate. You may use a single train/validation/test split, walk-forward expanding windows, time-series cross-validation, or any other scheme that respects causality. The scheme must be specified in your report and must be free of look-ahead at every stage.

If you take one principle from this section, take this one: the value of your alpha is bounded above by your worst look-ahead bias. A model that achieves a 5 percent Information Coefficient because it accidentally peeks at Close_t in a feature is worth nothing. The first thing the marker will check is whether every feature in your final model is provably observable at 15:50 ET on day t. Make that easy to verify.

5.3 Where to focus your creativity

Without telling you what to do, we can tell you where the interesting choices are. The strongest submissions tend to differentiate themselves on three dimensions, and we encourage you to invest your effort there:

- Feature design. The space of point-in-time features observable from daily data is much richer than it looks. Most of the difficult work is in encoding economic intuition into features that survive the no-intraday-information constraint.
- Target choice and conditioning. Predicting raw r_ON, t+1 is one option among many. A target that conditions on regime, on the magnitude of overnight moves, or on the dispersion of the day's intraday return often behaves better than the raw target.
- Robustness rather than peak performance. A model whose Sharpe is 1.3 and stable across rolling sub-periods will mark above a model whose full-sample Sharpe is 1.8 but whose performance is concentrated in two years.

Section 5 — questions you must explicitly answer in your report

1. State your information set $\mathcal{F}$_t and your target. List every feature with its formula and verify, line by line, that it is observable at 15:50 ET on day t.
2. State your model class and your training scheme. Justify both relative to the constraints of the problem (cross-sectional, daily, point-in-time).
3. Report the cross-sectional Information Coefficient: its mean, its t-statistic, its stability across years and across regimes you choose to define.
4. Provide an ablation: what is the marginal contribution of each feature group? Removing your top feature should not collapse the Sharpe.
5. What does your model do badly? Identify at least two regimes or sub-periods where the IC is weak, and discuss why.

6. Step 5 · From ranking to portfolio, with realistic costs

6.1 Translating scores into positions

On each date t, the scores s_{i,t} produced in Section 5 must be converted into actual dollar positions held overnight. The conventional approach is a top-and-bottom decile (or quantile) split: take the highest-scoring fraction of the eligible universe as the long basket, the lowest-scoring fraction as the short basket, and size each basket so that gross exposure equals 100 percent of the portfolio AUM (50 percent long, 50 percent short). Within a basket, you may choose equal weighting, score-weighted (proportional to s − median), or volatility-weighted. Justify your choice.

You must enforce dollar-neutrality at all times: the dollar value of the long book equals the dollar value of the short book on every date, after the participation cap of Section 3 has been applied. Beta-neutrality with respect to the market, while desirable, is not required for this brief; if you achieve it, document it as a robustness result. We do not require sector or industry neutrality, which is a meaningful additional layer of complexity that is out of scope for this coursework. If you choose to add it, report the marginal benefit explicitly; we expect it to be small.

6.2 Sizing under the participation cap

The participation cap of Section 3.3 imposes a maximum dollar position per name. Whenever an equal-weighted target position would exceed the cap, the position is reduced to the cap and the unused dollars are redistributed across the rest of the basket on a pro-rata basis. Iterate the redistribution until either no name is over the cap or all names in the basket are at the cap. If the basket cannot absorb the full target dollar exposure under the cap, gross exposure is reduced accordingly — do not artificially over-allocate to ineligible names just to hit a target gross.

6.3 The cost model — what you must charge

Your headline net Sharpe must be reported under the cost schedule of Table 6.1. Every leg of every round trip is charged at the rates in the table; the round trip is two legs (MOC entry plus MOO exit, or vice versa). Borrow is charged daily on the gross short notional at the rate determined by the borrow tier of the underlying name at each date.

Cost component	Rate	Applied to	Frequency
Commission, per leg	0.5 bps	Notional traded	Each leg
Auction slippage, per leg	1.5 bps	Notional traded	Each leg
Borrow — Tier A (General Collateral)	40 bps p.a.	Short notional	Daily
Borrow — Tier B (mid-tier specials)	200 bps p.a.	Short notional	Daily
Borrow — Tier C (deep specials)	800 bps p.a.	Short notional	Daily

Round-trip cost (commission + slippage), no borrow	**4.0 bps**	Notional turnover	Each round trip

Table 6.1 — The cost schedule applied to your back-test for headline results. These numbers are fixed; you do not choose them. Borrow tiers are assigned by your hard-to-borrow proxy of Section 4.

6.4 Tier-assignment rule for borrow

The borrow-tier assignment must be deterministic and reproducible from your hard-to-borrow proxy. Adopt the following rule for the headline numbers, with clear documentation if you deviate.

- Tier A (40 bps per annum) — any name not flagged hard-to-borrow at the date of the short.
- Tier B (200 bps per annum) — any name flagged hard-to-borrow under your proxy at a moderate-confidence threshold (you choose the threshold and document it).
- Tier C (800 bps per annum) — any name flagged hard-to-borrow under your proxy at a high-confidence threshold (you choose the threshold and document it).

Every short-position day is charged at the per-day equivalent of the annual rate for its tier. The exact daily charge is annual rate divided by 252. A short position of 1 million dollars held one night under Tier B incurs 200 bps × 1/252 × 1 million ≈ 79 dollars of borrow cost.

6.5 The deliverable performance numbers

Report your headline performance metrics at all three portfolio-AUM levels — 50 million, 250 million, and 1 billion dollars — in a single table. The metrics required are: net annualised return; net annualised volatility; net Sharpe ratio; maximum drawdown; daily turnover; gross exposure utilised on average; the fraction of days at gross-exposure cap; the borrow-cost contribution to gross-to-net Sharpe degradation.

In addition to the summary table, you must generate and submit a full QuantStats tear-sheet for the strategy at the 250-million-dollar AUM level, benchmarked against the S&P 500 Total Return index (SP500_TR) over the development window 2010 to end-2024. Use the open-source quantstats Python package and submit the rendered HTML report alongside your code repository. The tear-sheet must include all standard panels — cumulative return, drawdown, rolling Sharpe, monthly heat-map, distribution of daily returns, top-N drawdown periods, and the comparison metrics versus SP500_TR. This is the single artefact the marker will inspect first.

There is no minimum net Sharpe required to pass the coursework. The figure of 1.0 mentioned earlier in the brief is aspirational only — a number we suggest you push toward, not a threshold that determines whether your work is acceptable. What we mark on this dimension is the credibility of the number you report and the quality of the analysis you provide around it. A reproducible 0.6 with a sober diagnosis of where the alpha leaks is a much stronger submission than an optimistic 1.5 obtained through choices the marker cannot reproduce. Report whatever the methodology gives you, honestly, and explain it.

Section 6 — questions you must explicitly answer in your report

1. What are your basket sizes and weighting scheme inside each basket? What is the average daily turnover?
2. Provide the headline performance table at 50M, 250M, and 1B portfolio-AUM levels under the Section 6.3 cost schedule, computed on 2010–2024 only.

3. Decompose the gross-to-net Sharpe degradation: how many points come from commission, how many from slippage, how many from borrow?
4. Submit a QuantStats HTML tear-sheet of the 250M strategy benchmarked against SP500_TR over 2010–2024.
5. Conduct at least one stress-window analysis (e.g. 2020 Q1, late 2018, the 2022 drawdown). How does the strategy behave in dislocations?

7. Things to watch out for — questions the marker will ask

This section lists the questions a sceptical marker will put to your work. They are not bullet-point ornaments to add to a report; they are the substance against which a back-tested overnight strategy is judged. Answer them in your text directly, with evidence.

7.1 Look-ahead, in all its forms

The single most common reason an attractive-looking overnight strategy disappears under scrutiny is information leakage from the future into the present. The leakage takes many forms, some obvious and some subtle, and they are not all detected by a glance at the code.

- Are all your features observable at 15:50 ET on day t? In particular: features built from Close_t, High_t, or Low_t are not observable at 15:50 ET unless explicitly shifted by one trading day. Verify this for every feature.
- Is your earnings flag respecting the BMO/AMC convention of Section 2.1.2? An AMC announcement on day D is unknown at 15:50 ET on day D.
- Is your short-interest series respecting the publication lag plus the additional 2-day availability lag of Section 2.1.3?
- Are any cross-sectional standardisations (z-scores, ranks, deciles) computed using only data up to and including day t, or do they use future cross-sections?
- Are the parameters of any nonlinear feature transformation (winsorisation thresholds, breakpoints, etc.) calibrated on a strictly past sample, or have they been chosen on the full panel?
- Does your universe definition use any information not available at year-start? In particular: market capitalisation as of 31 December of year Y−1 is acceptable; market capitalisation as of 30 June of year Y is not, when defining the year-Y universe.

7.2 Statistical robustness

A back-test produces one realisation of a stochastic process. Strong claims about future performance require evidence that the realisation is representative, not anomalous.

- Is the Sharpe ratio you report stable across non-overlapping sub-periods? Provide a year-by-year breakdown.
- Are returns positively or negatively autocorrelated at the daily level? Adjust your reported Sharpe accordingly (Lo's autocorrelation correction is the standard reference).
- How does the strategy behave in the worst rolling 3-, 6-, and 12-month windows? Is the worst 12-month return tolerable for an investor?
- What fraction of the total return is concentrated in the top 5 percent of trading days? A strategy whose entire alpha is contained in 30 days out of 3 500 is not robust, even if the headline Sharpe is high.

7.3 Capacity and execution honesty

The back-test executes at the closing and opening auction prices recorded in your data. Real execution would not, in general, achieve those prices on positions of meaningful size.

- Have you applied the participation cap of Section 3.3 to the back-test? Re-running the strategy with no cap and a 5 percent cap should give different numbers; if they are identical, the cap is not biting and your strategy is implicitly assuming infinite liquidity.
- At each of the three portfolio-AUM levels you report, what is the average and peak per-stock dollar position? Is any single position above 5 percent of that name's ADV on any date?
- Is the slippage model of Section 3.3 reflected in the cost schedule of Section 6.3, or are the two specifications at different levels of pessimism? Document any reconciliation.

7.4 Borrow honesty

- Quantify, in basis points of gross return, the contribution of names with reported short interest above 10 percent of float. If that contribution is large, the strategy is implicitly relying on a part of the market where borrow risk is structural.
- How does the strategy behave if you replace your hard-to-borrow proxy with a hard exclusion of any name with short interest above some threshold? The two paths should give qualitatively similar Sharpes; if removing high-short-interest names destroys the alpha, the strategy is borrow-cost arbitrage in disguise.

7.5 Reporting integrity

- Is every chart in your report reproducible from a single command in your codebase? A chart whose generation is untraceable is not auditable.
- Is the seed of every randomised step fixed and reported? The same code on the same data must give the same numbers.
- Are the assumptions of Section 3.4 (portfolio AUM, basket size, participation cap) shown explicitly on every figure in which they bind?

8. Deliverables, format and grading

8.1 What you must submit

Your group will submit four artefacts, packaged as a single archive. Each is independently graded; together they constitute the complete deliverable for this coursework. There is no oral presentation.

- A written report of approximately 15 to 25 pages, in PDF format, addressed to a sceptical reader who knows the topic. The report must be self-contained and must answer, in its own narrative, every question explicitly listed in the boxed callouts at the end of Sections 2 to 6 and the bulleted questions of Section 7. The structure is yours; one workable layout is to follow the five-step structure of this brief.
- A QuantStats HTML tear-sheet of your final strategy at the 250-million-dollar AUM level, benchmarked against the S&P 500 Total Return index (SP500_TR), generated from the open-source quantstats Python package over the development window 2010 to end-2024. This is the single output the marker inspects first; it must therefore be reproducible from your repository under one command.
- A code repository that reproduces every number and every figure in the report and the tear-sheet. Submit either a Git URL (preferred) or a zipped archive. The repository must be fully compliant with the coding guidelines distributed alongside this brief — naming conventions, module layout, type-hint discipline, docstring format, test coverage, dependency management, and reproducibility constraints. Compliance is a marked dimension; non-compliance is penalised regardless of how well the strategy performs. Read the guidelines before writing your first line of code.
- A one-page innovation declaration that articulates, in your own words, what your group considers to be its added value relative to the large-language-model tools your group used during the work. Identify the two or three substantive choices — in problem framing, feature design, target construction, evaluation methodology, or any other dimension — where the human creativity of the group went beyond what an off-the-shelf LLM would have proposed. Be specific. A vague "we used the LLM as a coding assistant but we designed the strategy ourselves" is not a valid answer; we want to see the precise places where your judgment differentiated the work.

8.2 Format conventions

- Group size: maximum 3 students. Submissions from groups exceeding 3 students will not be accepted. Solo submissions are permitted but discouraged.
- All members of a group receive the same mark on the report and the code, with the possibility of a contribution-based adjustment based on the per-person commit history of the repository.
- All numbers in the report must be reproducible from the code repository under a single command. Numbers in the report that cannot be reproduced will be ignored when marking.
- All figures must include the relevant assumptions in the caption: portfolio AUM, basket size, participation cap, cost-schedule version. A figure without a caption disclosing the binding assumptions does not count toward the deliverable.
- The QuantStats tear-sheet must be the version generated from the master branch at the time of submission. A tear-sheet that does not reproduce from the submitted code is treated as not submitted.

- Citations are not required in the body of the report and a literature review is not part of the deliverable. An indicative bibliography is provided in Section 9; you may consult it but you are not asked to engage with it. Cite a paper only if you are using a specific number, definition, or method from it directly.
- The use of generative AI tools is permitted and expected. Beyond the one-page innovation declaration described above, do not embed AI-acknowledgement boilerplate in the body of the report — your own thinking is what the report is for.

8.3 How the work is graded

The total mark is 100 points, allocated across five rubric dimensions. The headline Sharpe ratio is one of five — it is important, but it is not the only thing being graded, and a high in-sample Sharpe cannot rescue a report that fails on the other four. The same code submitted will be re-run by the marker on a held-out window covering 2025 and 2026; held-out performance materially affects the empirical-performance and robustness dimensions of the rubric.

Dimension	Points	What the marker is looking for
Methodological soundness	**25**	Every feature point-in-time, every cost reflected, every assumption disclosed, no leakage from 2025–2026 into the development pipeline. The strongest single signal of a good submission.
Quality of reasoning, creativity and innovation declaration	**25**	Originality of modelling and design choices, clarity of trade-offs articulated, willingness to take an unconventional but defensible path. The one-page innovation declaration is read as part of this dimension.
Empirical performance — in-sample and held-out	**20**	Net performance is reported credibly under the Section 6.3 schedule on the 2010–2024 development window and on 2025–2026 when the marker re-runs your code. A reported Sharpe of order 1.0 is aspirational, not a pass/fail threshold; we mark the credibility and reproducibility of whatever number you report, the in-sample-to-held-out degradation, and the quality of the QuantStats tear-sheet.
Robustness and degradation analysis	**15**	Sub-period stability across 2010–2024, behaviour in 2020 Q1 and late 2022, sensitivity to feature ablation, sensitivity to cost-schedule perturbations, magnitude of in-sample-to-held-out degradation.
Code quality and guideline compliance	**15**	Strict compliance with the attached coding guidelines, single-command reproducibility, clear separation of pipeline stages, sensible repository structure, fixed random seeds, documented dependencies, deterministic regeneration of the QuantStats output.

Table 8.1 — Grading rubric. Total: 100 points. No oral component.

8.4 A final word on what we are looking for

This coursework is open-ended on purpose. Two equally capable groups working on the same brief will produce visibly different submissions, and that is the point. You are being assessed not on your ability to reproduce a known method but on your ability to take a well-defined economic question, choose a sensible path through it, defend your choices in writing, and produce numbers that someone else — including the marker, on the held-out window — could check.

The brief asks you to be creative inside a tightly specified set of constraints. The constraints — daily data only, market-neutral, US large-cap, 2010–2024 development with a held-out 2025–2026 evaluation, point-in-time everything, the cost schedule of Section 6.3, and the coding guidelines distributed with this brief — are non-negotiable. Within those, the design space is large, and the marks will go to the groups who explore it deliberately and document their reasoning honestly.

Good luck.

9. Indicative bibliography

The list below is provided as a starting reference set for groups who wish to read further. It is large on purpose — the field is large — and it is indicative, not prescriptive. None of the entries is required reading. The brief itself is self-contained and you can complete the coursework without opening any of these papers. If you do consult them, treat them as inspiration for your own reasoning, not as a recipe to imitate.

9.1 Overnight returns and the close-to-open premium

Akbas, F., Boehmer, E., Jiang, C., and Koch, P. D. (2022). Overnight returns, daytime reversals, and future stock returns. Journal of Financial Economics, 145(3), 850–875.

Bogousslavsky, V. (2021). The cross-section of intraday and overnight returns. Working paper.

Bondarenko, O., and Muravyev, D. (2023). Market return around the clock: A puzzle. Journal of Financial Economics.

Boyarchenko, N., Larsen, L. C., and Whelan, P. (2023). The overnight drift. Review of Financial Studies, 36(9), 3502–3547.

Cooper, M. J., Cliff, M. T., and Gulen, H. (2008). Return differences between trading and non-trading hours: Like night and day. SSRN working paper.

Hendershott, T., Livdan, D., and Rösch, D. (2020). Asset pricing: A tale of night and day. Journal of Financial Economics, 138(3), 635–662.

Knuteson, B. (2019). How to increase global wealth inequality for fun and profit. SSRN working paper.

Lachance, M.-E. (2023). Night trading: Lower risk but higher returns? Review of Financial Economics, 41, 347–363.

Lou, D., Polk, C., and Skouras, S. (2019). A tug of war: Overnight versus intraday expected returns. Journal of Financial Economics, 134(1), 192–213.

Qiao, K., and Dam, L. (2020). The overnight return puzzle and the T+1 trading rule in Chinese stock markets. Journal of Financial Markets, 50, 100534.

9.2 Microstructure, inventory and dealer behaviour

Bouchaud, J.-P., Bonart, J., Donier, J., and Gould, M. (2018). Trades, Quotes and Prices: Financial Markets Under the Microscope. Cambridge University Press.

Brunnermeier, M. K., and Pedersen, L. H. (2009). Market liquidity and funding liquidity. Review of Financial Studies, 22(6), 2201–2238.

Hasbrouck, J. (2007). Empirical Market Microstructure: The Institutions, Economics, and Econometrics of Securities Trading. Oxford University Press.

Madhavan, A. (2000). Market microstructure: A survey. Journal of Financial Markets, 3(3), 205–258.

Stoll, H. R. (2000). Friction. Journal of Finance, 55(4), 1479–1514.

9.3 Short selling, securities lending and hard-to-borrow

Beneish, M. D., Lee, C. M. C., and Nichols, D. C. (2015). In short supply: Short-sellers and stock returns. Journal of Accounting and Economics, 60(2–3), 33–57.

Boehmer, E., Jones, C. M., and Zhang, X. (2008). Which shorts are informed? Journal of Finance, 63(2), 491–527.

D'Avolio, G. (2002). The market for borrowing stock. Journal of Financial Economics, 66(2–3), 271–306.

Drechsler, I., and Drechsler, Q. F. (2014). The shorting premium and asset pricing anomalies. NBER working paper.

Engelberg, J. E., Reed, A. V., and Ringgenberg, M. C. (2018). Short-selling risk. Journal of Finance, 73(2), 755–786.

Geczy, C. C., Musto, D. K., and Reed, A. V. (2002). Stocks are special too: An analysis of the equity lending market. Journal of Financial Economics, 66(2–3), 241–269.

9.4 Market impact and execution

Almgren, R., and Chriss, N. (2001). Optimal execution of portfolio transactions. Journal of Risk, 3, 5–40.

Almgren, R., Thum, C., Hauptmann, E., and Li, H. (2005). Direct estimation of equity market impact. Risk, 18, 57–62.

Bouchaud, J.-P. (2010). Price impact. In Encyclopedia of Quantitative Finance, Wiley.

Donier, J., Bonart, J., Mastromatteo, I., and Bouchaud, J.-P. (2015). A fully consistent, minimal model for non-linear market impact. Quantitative Finance, 15(7), 1109–1121.

Frazzini, A., Israel, R., and Moskowitz, T. J. (2018). Trading costs. SSRN working paper.

Gârleanu, N., and Pedersen, L. H. (2013). Dynamic trading with predictable returns and transaction costs. Journal of Finance, 68(6), 2309–2340.

Gatheral, J. (2010). No-dynamic-arbitrage and market impact. Quantitative Finance, 10(7), 749–759.

9.5 Factor models and the cross-section of expected returns

Asness, C. S., Frazzini, A., and Pedersen, L. H. (2019). Quality minus junk. Review of Accounting Studies, 24(1), 34–112.

Carhart, M. M. (1997). On persistence in mutual fund performance. Journal of Finance, 52(1), 57–82.

Fama, E. F., and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. Journal of Financial Economics, 33(1), 3–56.

Fama, E. F., and French, K. R. (2015). A five-factor asset pricing model. Journal of Financial Economics, 116(1), 1–22.

Frazzini, A., and Pedersen, L. H. (2014). Betting against beta. Journal of Financial Economics, 111(1), 1–25.

Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. Review of Financial Studies, 28(3), 650–705.

9.6 Machine learning for cross-sectional return prediction

Chen, L., Pelger, M., and Zhu, J. (2024). Deep learning in asset pricing. Management Science, 70(2), 714–750.

Chinco, A., Clark-Joseph, A. D., and Ye, M. (2019). Sparse signals in the cross-section of returns. Journal of Finance, 74(1), 449–492.

Gu, S., Kelly, B. T., and Xiu, D. (2020). Empirical asset pricing via machine learning. Review of Financial Studies, 33(5), 2223–2273.

Israel, R., Kelly, B. T., and Moskowitz, T. J. (2020). Can machines "learn" finance? Journal of Investment Management, 18(2), 23–36.

Kelly, B. T., Pruitt, S., and Su, Y. (2019). Characteristics are covariances: A unified model of risk and return. Journal of Financial Economics, 134(3), 501–524.

López de Prado, M. (2018). Advances in Financial Machine Learning. Wiley.

9.7 Earnings, catalysts and event-driven effects

Bernard, V. L., and Thomas, J. K. (1989). Post-earnings-announcement drift: Delayed price response or risk premium? Journal of Accounting Research, 27, 1–36.

Engelberg, J. E., McLean, R. D., and Pontiff, J. (2018). Anomalies and news. Journal of Finance, 73(5), 1971–2001.

Frazzini, A., and Lamont, O. A. (2007). The earnings announcement premium and trading volume. NBER working paper.

Glasserman, P., Krstovski, K., Laliberte, P., and Mamaysky, H. (2025). Choosing news topics to explain stock-market returns. Journal of Forecasting, forthcoming.

9.8 Liquidity proxies and trading-cost measurement

Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. Journal of Financial Markets, 5(1), 31–56.

Corwin, S. A., and Schultz, P. (2012). A simple way to estimate bid-ask spreads from daily high and low prices. Journal of Finance, 67(2), 719–760.

Pástor, Ľ., and Stambaugh, R. F. (2003). Liquidity risk and expected stock returns. Journal of Political Economy, 111(3), 642–685.

Roll, R. (1984). A simple implicit measure of the effective bid-ask spread in an efficient market. Journal of Finance, 39(4), 1127–1139.

9.9 Volatility estimation and jumps

Andersen, T. G., Bollerslev, T., Diebold, F. X., and Labys, P. (2003). Modeling and forecasting realized volatility. Econometrica, 71(2), 579–625.

Garman, M. B., and Klass, M. J. (1980). On the estimation of security price volatilities from historical data. Journal of Business, 53(1), 67–78.

Lee, S. S., and Mykland, P. A. (2008). Jumps in financial markets: A new nonparametric test and jump dynamics. Review of Financial Studies, 21(6), 2535–2563.

Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. Journal of Business, 53(1), 61–65.

9.10 Backtesting, multiple testing and honest reporting

Bailey, D. H., Borwein, J., López de Prado, M., and Zhu, Q. J. (2014). Pseudo-mathematics and financial charlatanism: The effects of backtest overfitting on out-of-sample performance. Notices of the American Mathematical Society, 61(5), 458–471.

Harvey, C. R., Liu, Y., and Zhu, H. (2016). … and the cross-section of expected returns. Review of Financial Studies, 29(1), 5–68.

Ledoit, O., and Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. Journal of Empirical Finance, 15(5), 850–859.

Lo, A. W. (2002). The statistics of Sharpe ratios. Financial Analysts Journal, 58(4), 36–52.

Newey, W. K., and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. Econometrica, 55(3), 703–708.

9.11 Books and broad references

Cochrane, J. H. (2005). Asset Pricing (revised edition). Princeton University Press.

Grinold, R. C., and Kahn, R. N. (2000). Active Portfolio Management: A Quantitative Approach for Producing Superior Returns and Controlling Risk. McGraw-Hill.

Pedersen, L. H. (2015). Efficiently Inefficient: How Smart Money Invests and Market Prices Are Determined. Princeton University Press.

Appendix A — Data files provided for the project

The data files listed in this appendix are made available to all groups for the duration of the coursework. As a matter of principle, every file in the package is constructed from publicly available data or from publicly derivable transformations of public data — pricing and volume series from exchange tape, fundamentals from regulatory filings, calendar metadata from issuer disclosures, factor returns from cross-sectional regressions on public prices, and so on. No file relies on a private vendor feed, on intra-day microstructure information, or on a data source that an outside reader could not reconstruct in principle from the same public inputs.

All series are point-in-time within the conventions described in Sections 2.1.1 to 2.1.4 of this brief. The development window for the coursework is the start of 2010 to the end of 2024, regardless of the wider window over which a given file is supplied.

File	Contents	Frequency / window
prices.parquet	Daily OHLCV and market capitalisation per instrument; corporate-action-adjusted close and adjusted-close series.	Daily, 2000–2024
sp500_tr.parquet	S&P 500 Total Return index (SP500_TR), used as the benchmark for the QuantStats deliverable specified in Section 6.5.	Daily, 2000–2024
sp500_constituents.parquet	Historical S&P 500 index membership: per-instrument inclusion windows with effective add and remove dates. Use this for survivorship-bias-free universe construction and for backing out the eligible set on any historical date.	Continuous, 2000–2024
sp400_constituents.parquet	Historical S&P 400 (mid-cap) index membership on the same construction basis as the S&P 500 file above. Useful if you wish to extend the universe beyond large-cap or to study the large-cap / mid-cap boundary.	Continuous, 2000–2024
all_data.parquet	Wide panel of approximately 47 fundamental and technical features per instrument-day, derived from filings and price series.	Daily, 2010–2024
cheapness_scores.parquet	Four valuation pillars (price-to-earnings, price-to-book, EV/EBITDA, free-cash-flow yield) plus a composite	Daily, 2013–2024

	cheapness score and a regime-break flag.	
piotrosky.parquet	Piotroski F-score per instrument, normalised to a 0–9 scale, recomputed from the most recent quarterly fundamentals.	Quarterly, 2010–2024
earnings_calendar.parquet	Per-instrument earnings reporting dates with the BMO / AMC / Same-Day timing flag described in Section 2.1.2.	Continuous, 2010–2024
earnings_transfo.parquet	Earnings-derived features: standardised unexpected earnings (SUE), forward-EPS revision deltas (deps), and one- and four-quarter analyst-revision proxies (reps1, repsf4).	Continuous, 2010–2024
short_interest_transfo.parquet	Short-interest features: short-interest ratio (dsi), days-to-cover (dtcn), and the change in days-to-cover (ddtcn). Reflects the publication and availability lag described in Section 2.1.3.	Bi-monthly, 2010–2024
gics_info.parquet	GICS sector and industry classification per instrument. Static at provision; you are expected to be aware of M&A and reclassification events that this snapshot does not capture.	Static (~1 600 IDs)
regime.parquet	Daily macro-regime label per market: Underweight / Neutral / Overweight. Derived from publicly observable macro variables; the labelling rule is reproducible from public inputs.	Daily, 2016–2024
rolling_scores_downgrade.csv	Per-ticker probability of a credit-rating downgrade in the next 30 days, derived from public ratings and fundamentals.	Bi-weekly, 2010–2024
rolling_scores_upgrade.csv	Per-ticker probability of a credit-rating upgrade in the next 30 days, on the same construction basis as the downgrade file above.	Bi-weekly, 2010–2024

Table A.1 — Data files distributed with this brief. All files are constructed from publicly available data or from publicly derivable transformations of public data.

A.1 What this means in practice for your work

You may use any subset of these files. You are not required to use all of them. You may also augment them with additional public data you assemble yourself — for example, calendar effects, macroeconomic series from the FRED database, additional public-domain factor returns — provided your augmentations are documented in the report and reproducible from the code repository under the same single command as the rest of the pipeline.

You may not introduce data that is not publicly derivable. In particular, intraday quote or trade data, proprietary order-flow signals, broker-internal short-locate or borrow-rate feeds, and any vendor data product whose licence terms are not satisfied by your group's institutional access fall outside the permitted set. If you are uncertain whether a candidate data source qualifies, ask before using it; do not include it without authorisation.

The files above are sufficient to design a credible strategy that engages seriously with the brief. They are not optimal, and part of the assessment is your judgement about which subset to use, how to combine them, and which to ignore.